

The Dwarf Pilot, in plain words

The same four-expert verdict, retold in RAJA's voice — substance kept, jargon unpacked. The technical version lives beside this one; this is the read-it-straight-through pass.

Every fact RAJA leans on was re-checked against source: RAM 1024 MB, 500 ms interval, default `substrateSpeed 2`, the 128-frame streaming threshold, 4096-byte pages, and `MADV_NOHUGEPAGE` across every kernel family — all confirmed. Every separability number is the design's own *prediction*; no capture has run, so those are unverified as measurements by definition.

THE SPINE — ONE SENTENCE

The thesis' founding question — “can memory writes tell you which of 13 numerical dwarfs a VM is running” — is answered *no* on its own terms: the 500 ms snapshot blurs 13 dwarfs down to roughly 6–8 cells of a two-axis space, and the only honest thing left to test is whether a handful of compute-heavy kernels can be slowed enough to show their motion — before the campaign either earns its budget or collapses into a confusion matrix.

Why it read hard straight: four experts each argue in their own vocabulary (sampling theory, ML leakage, ECG morphology, memory-systems ground truth) and reach the *same* conclusions through different doors — so the real agreements were scattered across four voices instead of stated once.

1 The joint verdict

The target you were chasing does not exist; a smaller one does.

You were trying to recognize 13 dwarfs. The camera can't see them. The snapshot fires every 500 ms and ORs together every write in that half-second — a 2 Hz “stroboscope” — so thousands of compute passes smear into one flat image: “footprint fully lit.” Two of the four

experts, working alone, independently threw out both “dwarf” and “family” and landed on the same replacement: a point in a two-axis space.

- **Axis 1 — the always-visible still photo:** how big the footprint is, how saturated (dense-full vs sparse-stable), and the per-page magnitude regime (violent high-entropy re-seed vs quiet low-amplitude smoothing). About 3–5 shapes. You can always see this.
- **Axis 2 — the sometimes-visible motion:** does the footprint sit still, grow, shrink, slide as a band, or fade. You can only see this for a few kernels, and only if you deliberately enlarge them.

So the honest deliverable is: predict the *cell*, and hand back the **dwarf→cell confusion matrix** — which is exactly what `test_families_spec.md` asked for in the first place. The grand target quietly became the modest one the spec already wanted.

Motion survives only where the math outruns the memory.

Rhythm is dead — nothing periodic survives a 2 Hz camera. But a one-time *evolution* (LU's front shrinking, QR's front growing, a solver converging) is a slow drift, not a rhythm, and drift you can rescue — *if* you can stretch one pass across several snapshots without the array overflowing the guest's 1 GiB RAM. The governing quantity is arithmetic intensity: flops per byte written. Work must grow faster than space.

- **QR:** enlarge to ~2000, footprint ~7,600 pages, ~3% of RAM — cheap, do it first.
- **LU:** enlarge to ~3,800, ~27,000 pages, ~11% of RAM — moderate, do it second.
- **DP (edit-distance), stencil, FFT:** to slow them one snapshot you'd need 13.4 GB / 31.6 GB / 2.1 GB — infeasible on a 1 GiB guest, and for the stencil it's moot anyway.

The rescuable group is the dense $O(N^3)/O(N^2)$ fronts (QR, LU, and by the same law Cholesky, Floyd, Smith-Waterman). Everything $O(1)$ flops-per-byte (stencils, multigrid, nboddy, LBM, FDTD, DWT) plus the memory-bound DP/FFT is *permanently* aliased — it folds into a “fixed footprint, no motion” bucket whose only signal is size, plus magnitude-decay for the converging ones.

Buried caveat worth lifting: the QR/LU targets come from cache-resident smoke timing. At $N \approx 2000\text{--}4000$ the working set spills out of L2/L3, so each pass actually runs *slower* than N^3 predicts — meaning the true resize N is *smaller* and the rescue is *stronger* than the table says. But that's an estimate, not a measurement — Gate 1 exists to check it before any capture budget is spent.

The features: keep the placement-blind core, gate the address-shape extension.

- **Demoted:** the founding idea — treating each page's bytes as a signal and running FFT / wavelet / cepstrum on it. The stroboscope aliases it, and default `substrateSpeed 2` switches those heavy channels off. ECG even withdrew his own multigrid-cepstrum: cepstrum is the wrong tool for a monotone ramp — there's no repeating quefrency peak, so use ramp slope/curvature instead.
- **What survives (all four agree):** (1) per-page magnitude distribution (hamming/l1/l2) — separates the quiet writers (n-body, barnes-hut, MD, gibbs, belief-prop) from the violent re-seeders, and is *exactly* what APF's binary changed/not-changed indicator throws away; (2) footprint trajectory, for resized high-intensity fronts only; (3) magnitude-envelope decay, for converging solvers whose later writes flip fewer bits.
- **Model:** no autoencoder bank — it went degenerate on near-flat inputs in B1 and a forest beat it 19–59× on the same inputs. Use discriminative trees on a small feature vector.

The counting trap: effective n = number of *dwarfs*, not windows. A trajectory is one realization per cell; the windows inside it are points on one curve, not independent samples. A real generalization campaign therefore needs ~100–135 cells, not thousands of windows.

Schema catch: the existing `change_centroid/span/runs` channels are *per-page* (byte offsets inside one 4 KiB page), *not* whole-dump position. Any address-space-shape feature has to be rebuilt from the `page_index` column — and that column is the one most exposed to the next threat.

The deepest threat is free to test.

Kernels get their smooth address range from one `mmap()`, but `pmemsave` dumps guest-*physical* RAM, and the guest's buddy allocator decides where those pages actually land. Worse: every kernel calls `MADV_NOHUGEPAGE`, deliberately opting out of huge pages — which removes the one mechanism that would have forced physical contiguity. So the risk that changed pages scatter is *higher*, not lower. But the diff already emits `page_index`, so the test costs nothing: capture `gemm` at default for ~15 s at `--speed 0 --sparse`, take two consecutive snapshots deep in the run, and see whether the changed pages form a contiguous run or scatter across the 262,144-page range. This one check gates *every* address-shape claim in the pilot.

The leakage guards — what B1 taught them the hard way.

B1 got a hollow win (LORO leaked: 0.836 vs 0.457), so:

- Split by **dwarf**, never by cell or rep, for any cross-motif claim.
- The IDLE controls (spm/bfs/mc_pi) are the tripwire: they must stay mutually inseparable at floor after masking. If they separate, you're reading the environment, not the workload.
- Separation must survive removing absolute write volume — otherwise you just rebuilt APF.
- For trajectories: normalize to trace-*fraction* (kills the duration confound) and **drop warm-up snapshots** (the allocation / page-fault-in transient is itself a per-workload fingerprint).
- Normalize the capture by pass-duration, not by dimension, so footprint size can't sneak in as the label.

The pilot is one gated, cheapest-first afternoon.

Gate 0 (arithmetic, done — the table). Gate 1 (server, seconds per kernel: run `kernel_qr_v2 --dim 2000` and `kernel_lu_v2 --dim 3800` standalone, read the factorization counter — confirms the cache caveat before spending capture budget). Gate 2 (the contiguity capture — *the single most important go/no-go in the whole pilot*). Gate 3, only if Gate 2 passes: a ≥ 70 s capture at `--speed 0` (145 frames, clears the 128-frame streaming threshold) running resized QR+LU (front migration via Jaccard overlap of consecutive changed-page sets: overlap < 1 with a shifting boundary = front survived; ≈ 1 = still aliased), gemm as the static null, barnes_hut vs nbody as the address-shape pair, gibbs as the thinnest control (256 pages, magnitude-decay), stencil_jacobi as ECG's falsification control (the law says it stays saturated and static — if it separates, the law is wrong), dp cut to `--dim 2048` (not the coded 8192 — backpressure risk), plus rmat + lexer so QR can be held out leave-one-dwarf-out.

GO resized LU/QR separate from GEMM on trajectory *and* barnes_hut/gibbs separate on magnitude *and* it all survives level-normalization, placement-invariant-only features, and warm-up dropping → commit the ~100-cell campaign.

NO-GO LU/QR still collapse into GEMM → the “front” was memorization, temporal is dead even resized, and Part 2 honestly reduces to a spatial magnitude + footprint-size classifier — deliver the confusion matrix and stop paying for the temporal arm.

The one thing they still disagree on: pilot speed.

ECG wants speed 2 (his surviving leads live there, and it matches production cadence). Memory wants speed 0 (the rich magnitude channels, the contiguity test, and a free byte-FFT A/B all need it; speed 2 is a documented throughput compromise).

Resolution: run the *pilot* at speed 0 — cheap at pilot scale (only a few thousand changed pages), and it lets you observe axis-C, contiguity, and the byte-FFT A/B all at once — then confirm axis-C still survives at speed 2 before the campaign runs at production cadence. DSP's “free speed-0 A/B on the same retained dumps” is the bridge between the two votes.

2 The four solo verdicts

Each expert's final position after the cross-examination, in his own idiom.

DSP engineer — the stroboscope

What he sees. The whole system is one filter: a 500 ms boxcar-OR, a 2 Hz stroboscope. At default sizes it smears every kernel's internal geometry into “footprint fully lit,” so the visible dwarfs collapse into ~5 write-shape archetypes — a small grid of footprint-size $\times$ magnitude buckets. Rhythm is dead, and ML is right about that. But the *evolution envelope* (LU shrinking, QR growing) is a DC-to-0.05 Hz trend, not a rhythm, and trends you can rescue. His one governing rule: to stretch one evolution across many snapshots while still fitting 1 GiB, **work must grow faster than space**. Dense linear algebra ($O(N^3)/O(N^2)$) qualifies; anything where work $\approx$ space folds.

What he gave up. He adopted ML's placement-invariance as a hard filter and *withdrew his own “contiguous-vs-scatter shape” feature* — ASLR plus guest-physical fragmentation make absolute position non-reproducible. What survives are translation-invariant stats (footprint size, run-length histogram) and temporal derivatives of spatial moments (centroid velocity, span growth-rate), which don't care where the random origin sits. He also corrected ECG: read convergence-decay on `l1/l2`, not `hamming` — a $1e-10$ update still flips low mantissa bits, so hamming won't decay. And a converging solver shrinks its footprint *and* decays its magnitude together,

which is what distinguishes it from an LU shrink at constant front magnitude — the joint (size(t), l2(t)) curve is the real discriminator.

His pilot. Two arms, both speed 2, plus a free speed-o A/B on retained dumps. Arm A the collapse null (GEMM, stencil, hmm, diffusion — if they don't separate, archetype is the ceiling, ship the confusion matrix). Arm B the evolution rescue (GEMM/LU/QR resized + a slowed stencil — if {static, shrink, grow, decay} separate, the sub-type is real).

ML engineer — leakage and effective-n

What he sees. Neither “family” nor “dwarf” is a coherent learnable target. The real target is a point in a three-axis product: A = instantaneous footprint density (~3 levels at 2 Hz — scatter mostly collapses into dense under the boxcar-OR), B = footprint trajectory, C = per-page magnitude and its evolution. Dwarf survives only as a *region* — about 8–15 cells, informative for the numerical dwarfs but never reaching 13 (Jacobi, Floyd, Gibbs share a cell).

The trap he owns. Axes B and C are *once-per-cell* features. One LU run is one shrinking curve; the windows inside it are points on that curve, not independent samples. So your effective sample size collapses from thousands of windows to one realization per cell, and a defensible cross-motif claim needs ≥ 3 dwarfs $\times \geq 3$ workloads $\times \geq 3$ reps ≈ 100 –135 cells. The target got richer and the data got thinner at the same moment.

This is the single most load-bearing sentence in the whole document, and it is filed as a “catch,” not the headline: *the tension between a richer target and thinner data — not whether the signal exists — is what the whole program lives or dies on.*

Two new leak guards. Normalize each trajectory to trace-fraction (kills the duration confound); drop warm-up snapshots (the allocation transient is a workload fingerprint — a trajectory-flavored replay of B1's floor-recognition failure). **His pilot:** a sized, gated growing-front *generalization triad* — QR, rmat, lexer (three different motifs the spec all labels SEQUENTIAL-GROW) + GEMM contrast + spmv idle-null, speed 2, leave-one-*dwarf*-out on the axis-B label. If the held-out growing dwarf is recognized across a motif it never trained on, surviving all three guards $\rightarrow$ GO; if it collapses $\rightarrow$

archetype is the ceiling, ship the confusion matrix. Flagged as unverified and gating: whether speed 2 actually emits per-page magnitude or only the binary indicator — if not, axis C requires speed 0.

ECG signal scientist — the arithmetic-intensity law

What he sees. The write-visible dwarfs do not carry 13 separable morphologies. After the stroboscope integrates thousands of sub-snapshot passes (his own count: 16,000–30,000 passes per snapshot at smoke sizes), the target collapses to a small set of cells on two axes — an always-present spatial image and a conditionally-present temporal trajectory. The founding byte-domain FFT/wavelet/cepstrum lens is demoted.

His law. Temporal rescue is governed by arithmetic intensity, flops per byte. High-intensity fronts ($O(N^3)/O(N^2)$: LU, QR, DP, Floyd, Cholesky, Smith-Waterman) can be sized so one pass spans ≥ 3 snapshots under the cap; low-intensity saturators (stencils, multigrid, nbody — all $O(1)$ flops/byte) never can, at any feasible size. He *withdrew his own Phase-1 multigrid-cepstrum* as impractical and conceded multigrid to ML. Cepstrum is the wrong tool for a monotone ramp anyway — use the change-centroid ramp slope/curvature; reserve cepstrum for the one genuine recurrence, the convergence-decay envelope, which he keeps.

His falsifiable move. Only the high-intensity fronts separate from GEMM on the footprint-trajectory lead; low-intensity kernels stay inseparable except by footprint size, at every feasible size. So the pilot must include a *negative control*: a maximally-resized stencil. His law predicts it stays aliased — if it separates, the intensity rule is wrong. Flagged discrepancy: in his smoke read, qr/lu/dp emit no pass counter and stencil shows 30,157/snap, so memory's “QR ~ 0.9 ” must come from a larger sizing — which is precisely the resizability curve, and exactly what Gate 1 resolves.

Volatile-memory engineer — the ground truth

What he sees. Aliasing isn't uniform — it's a per-kernel continuum spanning four orders of magnitude at the same cadence, which is what turns sizing-rescue into a

targeted question instead of a blanket “temporal is dead.” None of these dwarfs has a *volume* problem — they write $2 \times -555 \times$ the ~ 118 -page OS floor — but ~ 15 of ~ 20 collapse into the same footprint-magnitude bucket regardless of identity, and floor-subtraction does nothing to fix that. Speed 2 dropping the heavy channels is documented and deliberate (the `--help` says use speed 2–4 at 500 ms); the magnitude/texture leads need an explicit `--speed 0`.

His hard numbers. The per-kernel resize table turns ECG's law into RAM-bounded fact: QR ~ 2000 at $\sim 3\%$ RAM; LU ~ 3800 at $\sim 11\%$; DP 13.4 GB and stencil 31.6 GB, both infeasible on a 1 GiB guest. The cache-effect caveat makes QR/LU's real N smaller, strengthening the rescue — but Gate 1 must confirm. He also traced two mislabels to their source: `sddmm`'s sparsity lives in index metadata a byte-differ can't see (an inherent limit), while `fem_assembly` zeroes the whole matrix every pass (a fixable coding choice, not a law).

His gate. All 61 kernels opt out of transparent huge pages (`MADV_NOHUGEPAGE`), so physical contiguity is more exposed — but the differ already emits `page_index`, so a single ~ 15 s gemm capture at `--speed 0 --sparse` settles it before any address-shape feature is trusted. **Note:** proceed, gates cheapest-first (0 arithmetic $\rightarrow$ 1 server timing $\rightarrow$ 2 contiguity $\rightarrow$ 3 pilot at speed 0, ≥ 70 s). If Gate 2 shows severe scattering, drop `barnes_hut/nbody` and the LU/QR-front claims, keep only magnitude + footprint-size, and make contiguity its own follow-up study.

THE SHAPE OF IT

The honest result is already in hand — memory writes identify a *cell*, not a dwarf, and the confusion matrix is the guaranteed deliverable. Everything still open rides on one cheap afternoon: Gate 2's contiguity check decides whether any address-shape feature is real at all, and the resized-QR/LU vs GEMM split decides whether the temporal arm is worth a campaign. Urgent and free: run Gates 0–2. Everything downstream waits on Gate 2 passing.

RAJA retelling of `DWARF_PILOT_DESIGN.html` (the technical four-expert design review). Companion, not replacement — the technical version keeps the full tables and the verbatim verdicts. Design only; no capture has run, so every separability statement is a prediction the gated pilot exists to test.